

## SEMESTER S8

### COMPUTER GRAPHICS

|                                            |                |                    |                |
|--------------------------------------------|----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OECS835</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0        | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3              | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None           | <b>Course Type</b> | Theory         |

#### Course Objective:

1. To provide strong technological concepts in computer graphics including the three-dimensional environment representation in a computer, transformation of 2D/3D objects and basic mathematical techniques and algorithms used to build applications.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Contact Hours</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Basics of Computer graphics</b> - Basics of Computer Graphics and its applications. Video Display devices - LED, OLED, LCD, PDP and FED and reflective displays. Random and Raster scan displays and systems.<br><b>Line and Circle drawing Algorithms</b> - Line drawing algorithms- Bresenham's algorithm, Liang-Barsky Algorithm, Circle drawing algorithms - Midpoint Circle generation algorithm, Bresenham's Circle drawing algorithm. | <b>10</b>            |
| <b>2</b>          | <b>Geometric transformations</b> - 2D and 3D basic transformations - Translation, Rotation, Scaling, Reflection and Shearing, Matrix representations and homogeneous coordinates.<br><b>Filled Area Primitives</b> - Scan line polygon filling, Boundary filling and flood filling.                                                                                                                                                             | <b>10</b>            |
| <b>3</b>          | <b>Transformations and Clipping Algorithms</b> - Window to viewport transformation. Cohen Sutherland and Midpoint subdivision line clipping algorithms, Sutherland Hodgeman and Weiler Atherton Polygon clipping algorithms.                                                                                                                                                                                                                    | <b>8</b>             |
| <b>4</b>          | <b>Three dimensional graphics</b> - Three dimensional viewing pipeline. Projections- Parallel and Perspective projections. Visible surface detection algorithms- Back face detection, Depth buffer algorithm, Scan line algorithm, A buffer algorithm.                                                                                                                                                                                          | <b>8</b>             |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written) | Total |
|------------|-----------------------------|----------------------------------------|-----------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                      | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                      | Part B                                                                                                                                                                                                                                                                   | Total |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p>(8x3 =24 marks)</p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 subdivisions.</li> </ul> <p>(4x9 = 36 marks)</p> | 60    |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                        | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|------------------------------------------------------------------------|------------------------------------|
| CO1            | Understand the principles of computer graphics and displays            | K2                                 |
| CO2            | Illustrate line drawing, circle drawing and polygon filling algorithms | K3                                 |
| CO3            | Illustrate 2D and 3D basic transformations and matrix representation   | K3                                 |
| CO4            | Demonstrate different clipping algorithms and 3D viewing pipeline.     | K3                                 |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| CO2 | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |
| CO3 | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |
| CO4 | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                    |                                                     |                              |                         |
|-------------------|----------------------------------------------------|-----------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                           | <b>Name of the Author/s</b>                         | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| <b>1</b>          | Computer Graphics : Algorithms and Implementations | D. P. Mukherjee, Debasish Jana                      | PHI                          | 1/e, 2010               |
| <b>2</b>          | Computer Graphics with OpenGL                      | Donald Hearn, M. Pauline Baker and Warren Carithers | PHI                          | 4/e, 2013               |

| <b>Reference Books</b> |                                             |                                                    |                              |                         |
|------------------------|---------------------------------------------|----------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                    | <b>Name of the Author/s</b>                        | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| <b>1</b>               | Introduction to Flat Panel Displays         | Jiun-Haw Lee, I-Chun Cheng, Hong Hua, Shin-Tson Wu | Wiley                        | 1/e, 2020               |
| <b>2</b>               | Computer Graphics and Multimedia            | ITL ESL                                            | Pearson                      | 1/e, 2013               |
| <b>3</b>               | Computer Graphics                           | Zhigang Xiang and Roy Plastock                     | McGraw Hill                  | 2/e, 2000               |
| <b>4</b>               | Principles of Interactive Computer Graphics | William M. Newman and Robert F. Sproull            | McGraw Hill                  | 1/e, 2001               |
| <b>5</b>               | Procedural Elements for Computer Graphics   | David F. Rogers                                    | McGraw Hill                  | 1/e, 2017               |
| <b>6</b>               | Computer Graphics                           | Donald D Hearn, M Pauline Baker                    | Pearson                      | 2/e, 2002               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                                                                                           |
| <b>1.</b>                             | Computer Graphics By Prof. Samit Bhattacharya at IIT Guwahati<br><a href="https://onlinecourses.nptel.ac.in/noc20_cs90/preview">https://onlinecourses.nptel.ac.in/noc20_cs90/preview</a> |